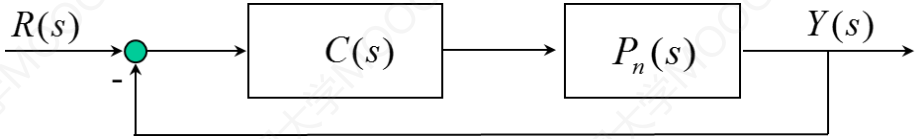
Analysis of Feedback Systems
– Relative Stability and Robust Stability
(反馈系统分析 – 相对稳定性和鲁棒稳定性)

Relative Stability and Margins

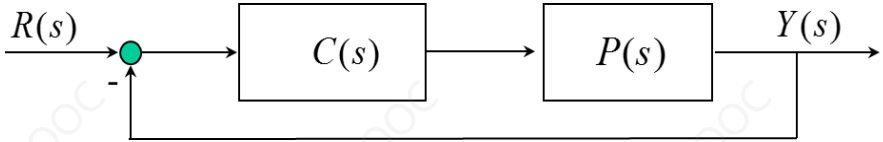
- **Relative Stability (相对稳定性)**
  - Traditional controller design is based on, $P_n(s)$, the nominal model of a physical system **(传统控制器设计是基于物理系统的名义模型)**
  - In reality, the actual model $P(s)$ often differs from the nominal model **(实际上,真实模型经常与名义模型有所不同)**
  - It is thus important to have an idea on the degree of model imperfection that the nominal controller design can tolerate without losing actual closed-loop stability, i.e., relative stability **(名义控制器在没有丧失实际闭环系统稳定性的情况下能承受模型缺陷的程度是非常重要的，即相对稳定性)**
  - Traditional quantities in measuring relative stability are gain and phase margins. A better modern version will be the sensitivity peak or the robust stability theorem introduced in the following **(衡量相对稳定性的传统量化指标是幅值裕度和相位裕度。更好的现代量化指标是以下将要介绍的灵敏度峰值或鲁棒稳定性理论)**

Relative Stability and Margins

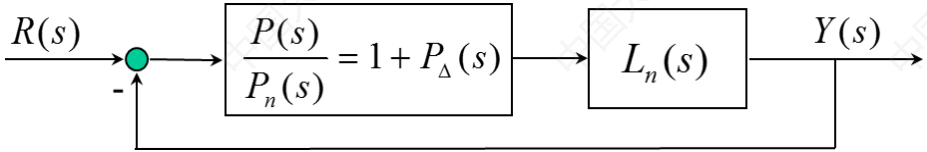
- **Feedback Controller Design Model:**



- **Actual Closed-Loop System**



$\Updownarrow$



- nominal TF of the plant (系统名义传递函数): $P_n(s)$

- nominal open-loop TF (系统名义开环传递函数): $L_n(s)$

- additive modeling error (AME) (加性建模误差): $\tilde{P}(s) = P(s) - P_n(s)$

- multiplicative modeling error (MME) (乘性建模误差):

$$P_\Delta(s) = \left(P(s) - P_n(s)\right) / P_n(s)$$

- actual open-loop TF (实际开环传递函数): $L(s) = \tilde{P}(s) L_n(s)$

Q: Suppose $L_n(s)$ is stable and the nominal CL system (i.e., $P_\Delta(s)=0$) is stable, when will the actual CL system become unstable? **(假设$L_n(s)$是稳定的且名义闭环系统式稳定的,实际开换闭环系统什么时候会不稳定 ?)**

Relative Stability and Margins

- **Gain Margin (幅值裕度)**
  - The change in *static* gain of open loop TF $L(s)$ required to make the CL system unstable **(使闭环系统不稳定的开环传函所需的静态增益改变量)**
  - Systems with greater gain margin M_g can withstand greater changes in system gain parameters before becoming unstable in closed loop **(在闭环系统变得不稳定前,更大的幅值裕度能承受更大的系统增益参数变化)**

when $P(s) = K_p P_n(s)$

$$\Rightarrow L(s) = C(s)P(s) = C(s)K_p P_n(s) = K_p L_n(s)$$

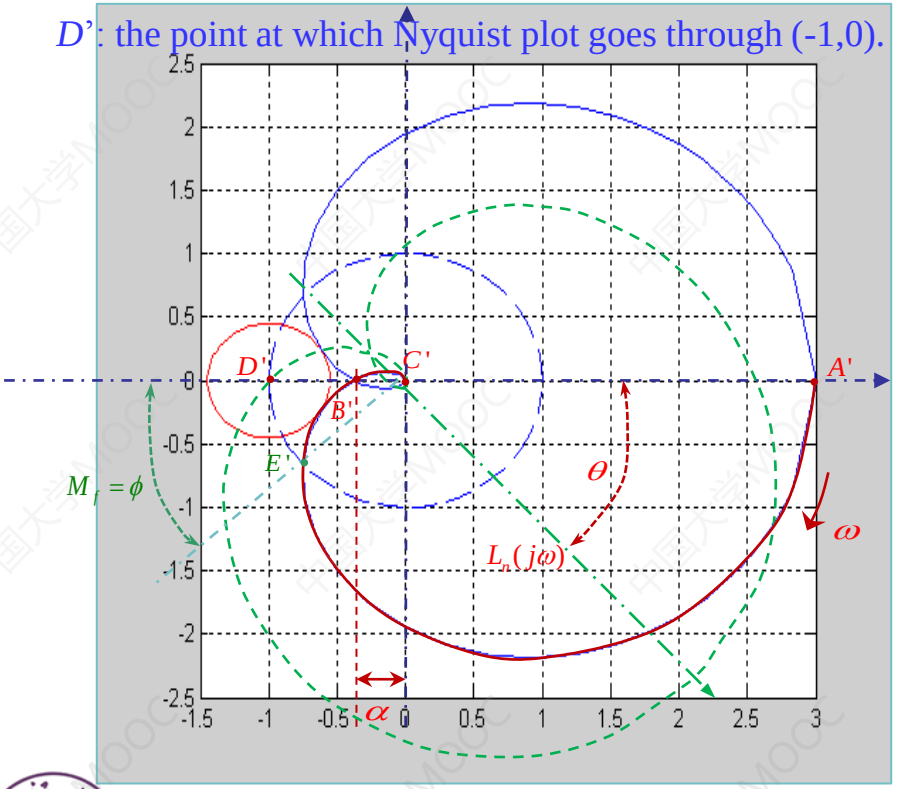
- **Phase Margin (相位裕度)**
  - The change in *pure* phase shift of open loop TF $L(s)$ required to make a closed loop system unstable **(使闭环系统不稳定的开环传函所需的纯相位变化量)**
  - Systems with greater phase margin M_f can withstand greater time-delay effect **(更大相位裕度的系统能承受更大的时延影响)**

$L(s) = K_p L_n(s)$ where $|L(s)| = |L_n(s)|$ and

$$\angle L(j\omega) = \angle L_n(j\omega) - \theta$$

Stability Margins via Nyquist Criterion

D': the point at which Nyquist plot goes through (-1,0).



$$L_n(s) = \frac{3}{(s+1)^3}$$

***N*=0, *P*=0,→ *Z*=*N*+*P*=0 → Nominal CL is stable.**

- **Effect of pure gain:**

For $L(s)=K_pL_n(s)$, *P* does not change,

then, the actual CL system is stale if *N* does not change.

→ As long as $K_p|\alpha|<1$, then *N* will not change.

→ Actual CL system is stable: $K_p < \frac{1}{\alpha} = M_g$

- **Effect of phase lag:** $L(s)=e^{-\tau s}L_n(s)=e^{-j\theta}L_n(s)$

→ $|L(s)|=|L_n(s)|,\ \angle L(s)=\angle L_n(s)-\theta$

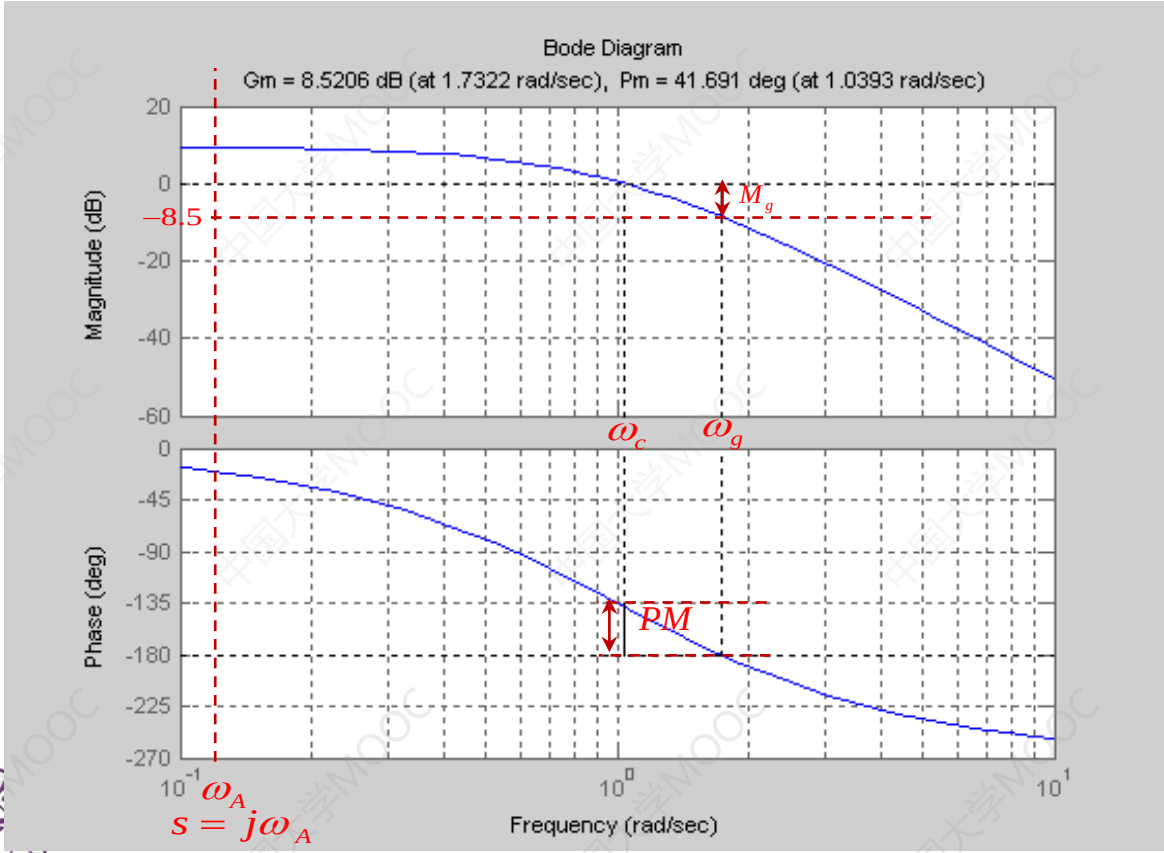
→ As long as $\theta < \phi = M_f$, *N* will not change

→ CL stability will not change.

B': the point at which Nyquist plot goes through the real axis.

E': the point at which Nyquist plot goes through the unit circle.

Stability Margins via Bode Plots

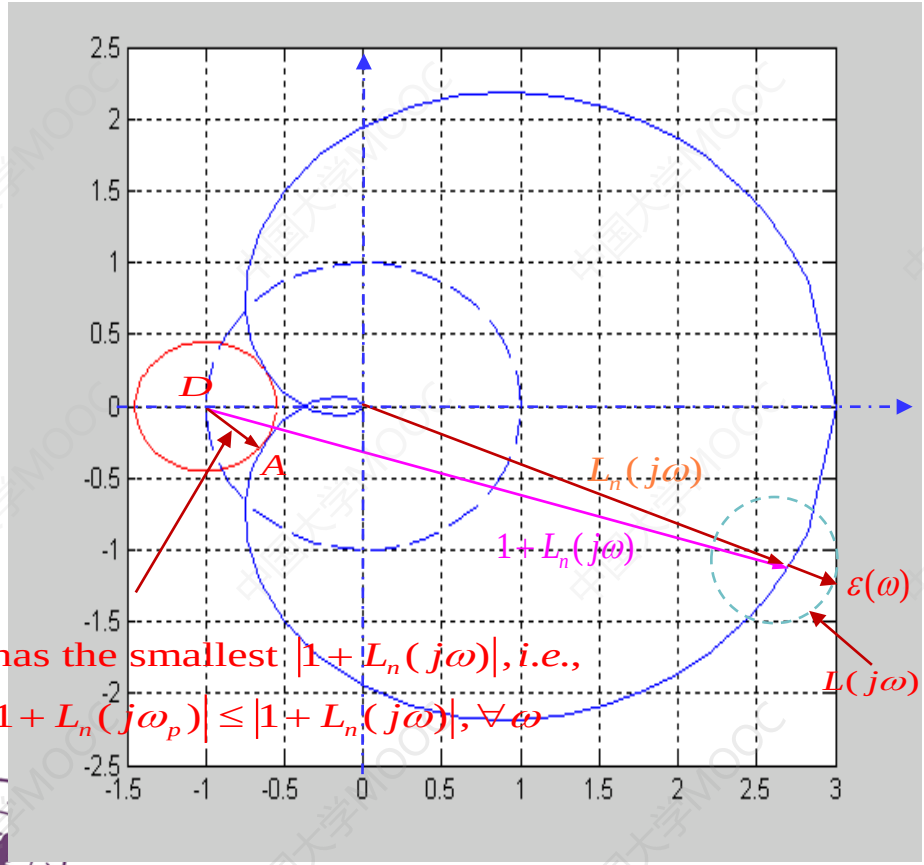


$$L_n(s) = \frac{3}{(s+1)^3}$$

$$M_g = 8.5dB \approx 2.5$$

$$PM = 41.69^{\circ}$$

Sensitivity Peak via Nyquist Criterion



has the smallest $|1+L_n(j\omega)|$, *i.e.*,

$$|1+L_n(j\omega_p)| \le |1+L_n(j\omega)|, \forall \omega$$

$L(j\omega)$ will lie inside the disk

$$L_n(s) = \frac{3}{(s+1)^3}$$

$$L(s) = L_n(s) + \tilde{L}(\omega)$$

$$\Rightarrow L(j\omega) = L_n(j\omega) + \tilde{L}(j\omega), \quad \text{Assume } |\tilde{L}(j\omega)| < \varepsilon(\omega)$$

As long as $\forall \omega,\ \varepsilon(\omega) < |1+L_n(j\omega)|$

then $L(s)$ will have the same N as $L_n(s)$

→ Stability will not change.

Sensitivity Peak

- **Sensitivity function reaches its peak at ω_p, the frequency having the shortest distance η to the -1 point on the Nyquist plot (灵敏度函数在ω_p处达到峰值点, 这个频率在奈奎氏特曲线上具有离-1最短的距离η)**

$$\left|S_n\left(j\omega_p\right)\right| = \frac{1}{\eta} \geq \left|S_n\left(j\omega\right)\right|, \forall\omega, \Longrightarrow \text{As } \eta = \frac{1}{\left|S_n\left(j\omega_p\right)\right|} = \left|1+L_n(j\omega_p)\right| \leq \left|1+L_n(j\omega)\right|, \ \forall\omega$$

  - Dealing with the change in *frequency* gain of open loop TF $L(s)$ required to make the CL system unstable; the change in frequency gain $|L(j\omega)|$ can be caused by the change in both static gain and phase shift **(引起闭环系统不稳定的开环传函在频域上的增益变化量，该增益变化量可由静态增益和相位变化共同造成)**
  - The larger the sensitivity peak is, the closer the loop will be to instability **(灵敏度峰值越大，闭环系统越接近不稳定)**
  - A system may have good gain and phase margins, and yet a very large sensitivity peak indicating a very delicate stability condition. The converse is not true: sensitivity performance implies minimal values for gain and phase margins as given by **(系统可能具有较好的幅值裕度和相位裕度，但具有较大的灵明度峰值，意味着系统的稳定条件非常敏感。反之不成立：灵敏度性能意味着幅值裕度和相位裕度的最小值)**

$$M_f \geq 2\arcsin\left(\frac{1}{2}\eta\right)$$

Robust Stability

- **Robust Stability Theorem (鲁棒稳定性理论)**
  - A plant with a nominal TF of $P_n(s)$ and a true TF of $P(s)$ **(系统具有名义传函$P_n(s)$和实际传函$P(s)$)**.
  - Assume that a controller $C(s)$ designed to achieve nominal internal stability (i.e., no unstable pole/zero cancellation and $L_n(s)= P_n(s)C(s)$ is stable), and $L_n(s)= P_n(s)C(s)$ and $L(s)= P(s)C(s)$ have the same number of unstable poles **(设计控制器$C(s)$使得名义系统达到内部稳定性(也就是没有不稳定零极点相消且$L_n(s)$是稳定的), $L_n(s)$和$L(s)$具有相同的不稳定极点)**.
  - Then, a sufficient condition for stability of the actual feedback loop by applying the same controller to the true plant is that **(那么，将相同控制器作用于实际系统，其反馈系统稳定性的充分条件是)**

$$\left|T_n(j\omega)\right|\left|P_\Delta(j\omega)\right| = \left|\frac{P_n(j\omega)C(j\omega)}{1+P_n(j\omega)C(j\omega)}\right|\left|P_\Delta(j\omega)\right| < 1$$

where $P_\Delta(j\omega)$ is the frequency response of the multiplicative modeling error **($P_\Delta(j\omega)$是乘性建模误差的频率响应)**

Robust Stability

- **Proof of Robust Stability Theorem (鲁棒稳定性的证明)**

Proof: $P(s) = P_n(s) + \tilde{P}(s) = P_n(s)(1 + P_\Delta(s))$

$$\Rightarrow L(s) = P_n(s)[1 + P_\Delta(s)]C(s) = L_n(s)(1 + P_\Delta(s)) = L_n(s) + \underbrace{L_n(s)P_\Delta(s)}_{\tilde{L}(s)}$$

$\Rightarrow$ since actual CL is stable if

$$L_n(j\omega)P_\Delta(j\omega) = \varepsilon(\omega) < |1 + L_n(j\omega)|, \quad \forall j\omega$$

$$\Rightarrow |P_n(j\omega)C(j\omega)P_\Delta(j\omega)| < |1 + P_n(j\omega)C(j\omega)|$$

$$\Rightarrow \frac{|P_n(j\omega)C(j\omega)|}{|1 + P_n(j\omega)C(j\omega)|}|P_\Delta(j\omega)| = |T_n(j\omega)||P_\Delta(j\omega)| < 1$$

The theorem indicates that for the frequency range where the system has large extent of modeling error, for robust stability, the nominal sensitivity function $S_n(j\omega)$ should be close to 1, i.e., almost open-loop control ! **(理论表明在系统具有较大建模误差的频段，为实现鲁棒稳定性，名义灵敏度函数必须要接近于1，也就是近似开环控制)**

Robust Stability Example

Nominal System:

$$P_n(s)C(s) = \frac{0.1}{s(s+1)^2}$$

True Plant:

$$P(s) = e^{-s\tau}P_n(s)$$

MME:

$$P_\Delta(s) = e^{-s\tau} - 1$$

Q1: Find exact value of time delay that leads to closed-loop instability
(找出引起闭环系统不稳定的时延精确值)

$$L(s) = P(s)C(s) = e^{-s\tau}P_n(s)C(s) = e^{-s\tau}L_n(s)$$

$$\Rightarrow |L(j\omega)| = |e^{-j\omega\tau}||L_n(j\omega)| = |L_n(j\omega)|, \quad \angle L(j\omega) = \angle e^{-j\omega\tau} + \angle L_n(j\omega) = -\omega\tau + \angle L_n(j\omega)$$

At the stability boundary, L(s) Nyquist plot has to go through -1 points (在稳定边界上，*L(s)*乃奎氏特曲线穿过-1点). Thus,

$$\exists \omega_p, s.t., \quad L(j\omega_p) = -1 \Leftrightarrow \begin{cases} |L(j\omega_p)| = 1 \\ \angle L(j\omega_p) = -180^\circ \end{cases}$$

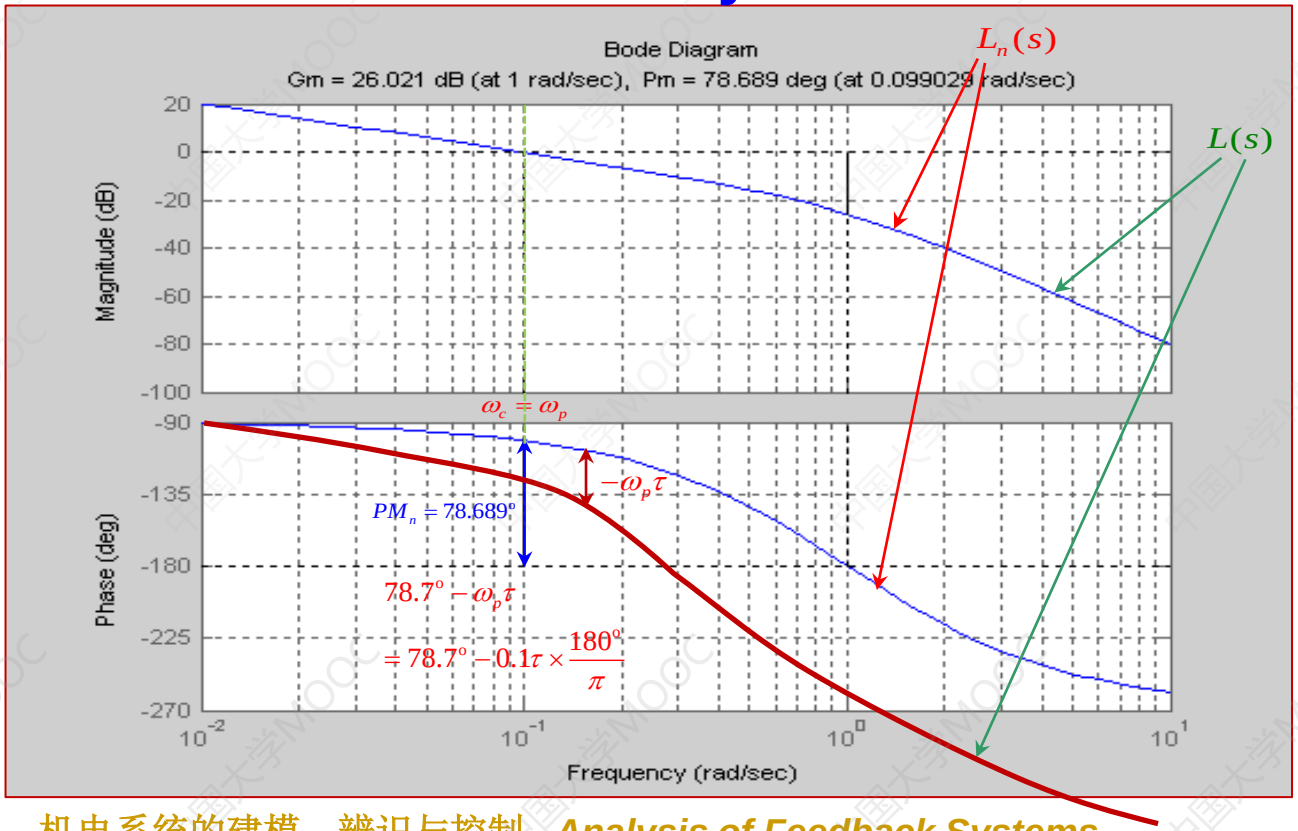
$$\Rightarrow |L_n(j\omega_p)| = 1 \Rightarrow \omega_p = \omega_c = 0.1\text{rad/sec}$$

$$\Rightarrow -180^\circ \le \angle L(j\omega_p) = -\omega_p\tau + \angle L_n(j\omega_p)$$

$$\frac{\pi}{180^\circ}(-180^\circ) \le -0.1\tau + \frac{\pi}{180^\circ}(-180^\circ + PM) \Rightarrow \tau \le \frac{\pi}{180^\circ} \times PM / 0.1$$

Robust Stability Example

- **Bode Plots of Nominal System**



$$-180^\circ \le \angle L(j\omega_p) = -\omega_p\tau + \angle L_n(j\omega_p)$$

$$= -0.1\tau + \frac{\pi}{180^\circ}(-180^\circ + 78.7^\circ)$$

$$\Rightarrow \tau \le \frac{\pi}{180^\circ} \times 78.7^\circ / 0.1 = 13.87\,\text{sec}$$

Robust Stability Example

Q2: Obtain an estimate for critical time delay using robust stability theorem
(运用鲁棒稳定理论获得临界时延的估计量)

$$\left|P_\Delta(j\omega)\right| = \left|e^{-j\omega\tau} - 1\right| = \left|\cos(\omega\tau) - j\sin(\omega\tau) - 1\right| = \sqrt{[\cos(\omega\tau)-1]^2 + [\sin(\omega\tau)]^2}$$

$$T_n(s) = \frac{L_n(s)}{1+L_n(s)} = \frac{0.1/[s(s+1)^2]}{1+0.1/[s(s+1)^2]} = \frac{0.1}{s(s+1)^2+0.1}$$

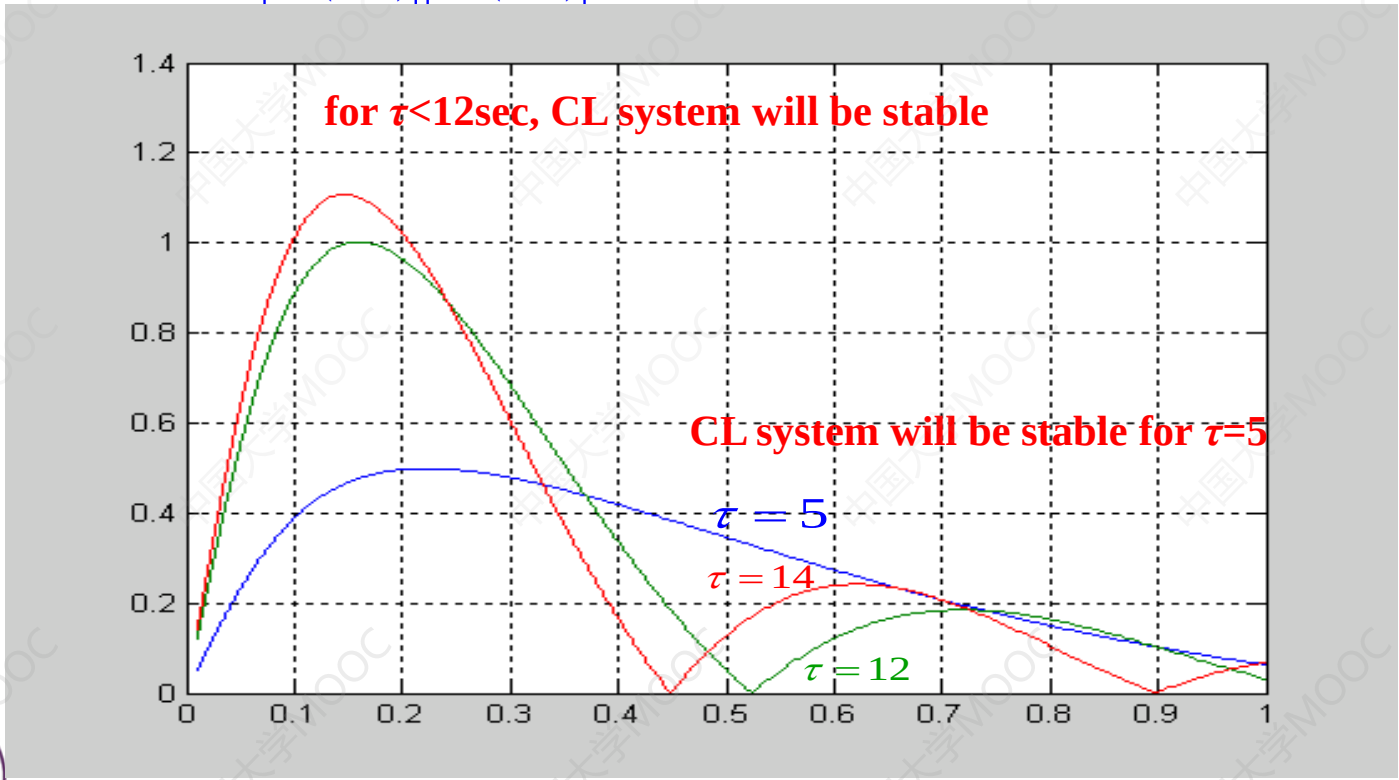
need to find $\tau_{\max}$ s.t.

$$\left|T_n(j\omega)\right|\left|P_\Delta(j\omega)\right| < 1$$

$$\forall\,\omega, \forall\,\tau < \tau_{\max}$$

Robust Stability Example

- **Plots of** $\left|T_n(j\omega)\right|\left|P_\Delta(j\omega)\right|$



Robust Stability Example

Q3: Comment on the difference if any (不同点)

- **Conservativeness of Robust Stability Theorem (鲁棒稳定性理论的保守性)**
  - A sufficient condition that gives the worst case requirement **(给出最坏情况需求的充分条件)**
  - Derived based on the magnitudes of MME and complementary sensitivity function **(基于乘性建模误差和互补灵敏度函数的幅值来推导)**

Actual and Nominal Sensitivity Functions

- **Actual Achieved Sensitivity Functions** (实际获得的灵敏度函数)

$$S(s) = S_n(s)S_\Delta(s)$$
$$T(s) = T_n(s)\big(1+P_\Delta(s)\big)S_\Delta(s)$$
$$S_i(s) = S_{in}(s)\big(1+P_\Delta(s)\big)S_\Delta(s)$$
$$S_u(s) = S_{un}(s)S_\Delta(s)$$

where

$$S_\Delta(s) = \frac{1}{1+T_n(s)P_\Delta(s)}, \quad \text{Error Sensitivity}$$

$$P_\Delta(s) = \frac{P(s)-P_n(s)}{P_n(s)}, \quad \text{MME}$$

$$L(s) = C(s)P(s) = C(s)P_n(s)[1+P_\Delta(s)]$$
$$= L_n(s) + C(s)P_n(s)P_\Delta(s)$$
$$S(s) = \frac{1}{1+L(s)} = \frac{1}{1+L_n(s)+L_n(s)P_\Delta(s)}$$
$$= \underbrace{\frac{1}{1+L_n(s)}}_{S_n(s)} \cdot \frac{1}{1+\underbrace{\frac{L_n(s)}{1+L_n(s)}}_{T_n(s)}P_\Delta(s)}$$

Summary

- **Relative Stability (相对稳定性)**
  - Empirical Stability Margins (经验稳定裕度)
    - Gain and phase margins, Sensitivity peaks (幅值和相位裕度，灵敏度峰值)
  - Robust Stability of Feedback System (反馈系统的鲁棒稳定性)
    - Robust Stability Theorem (鲁棒稳定理论)